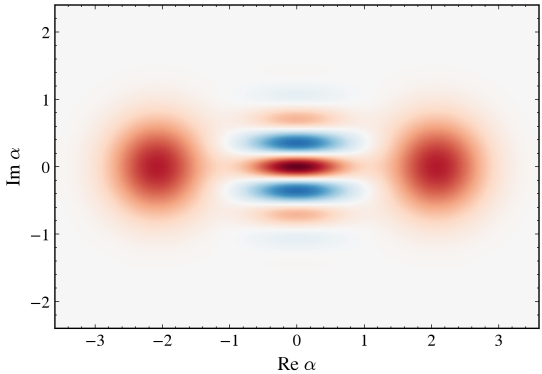
DUQUANTUM 2026 ALICE & BOB

Cat qubits, simulated

How a cat qubit is made, how well it keeps its bit, and how it is read

Team Shunya



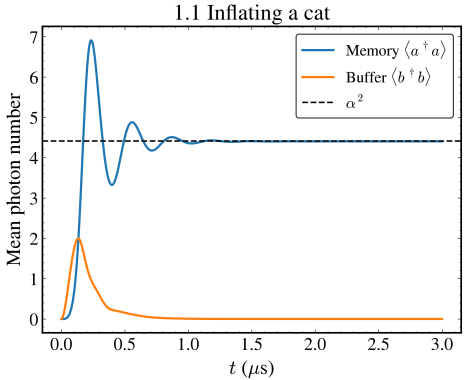
A cat state: one resonator oscillating two opposite ways at once

A cat qubit is protected against one of the two errors

Qubits make two kinds of mistakes: bit flips, where 0 turns into 1, and phase flips.

A cat qubit stores 0 and 1 as two opposite oscillations of one microwave resonator. A pump keeps pushing it back to those two states.

Bit flips become very rare, so only phase flips are left to correct.



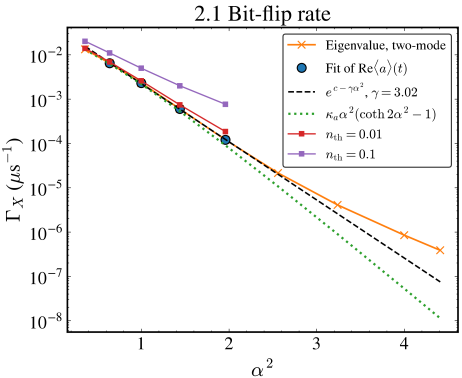
Switching on the pump fills the resonator to 4.4 photons within about 1 µs.

Bigger cats flip far less often

Each point is the bit-flip rate from one simulation, for a cat of a given size.

The line falls steeply: for small cats, each extra photon makes bit flips about 20 times rarer.

Phase flips grow only in proportion to the photon number, so a bigger cat is a good trade.



Bit-flip rate (log scale) against photon number

At our working point a bit flip takes seconds

2.6 s
average time to a bit flip

1.94 µs
average time to a phase flip

~ 10^6
how much rarer bit flips are

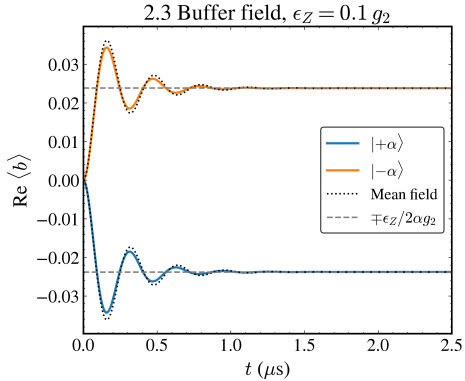
A cat of 4.4 photons, simulated with both resonators and no thermal noise

The partner resonator tells us which state the cat is in

A small push on the cat makes the partner's output go up for one state and down for the other. It settles within 1 μs.

The push also causes a few flips, so the best readout error is about 0.4%. Growing the cat during the readout cuts that error 13 to 18 times.

Code and every number: github.com/yubrajbhandari923/DuQuantum, branch ak-changes



Output signal for the two states over time